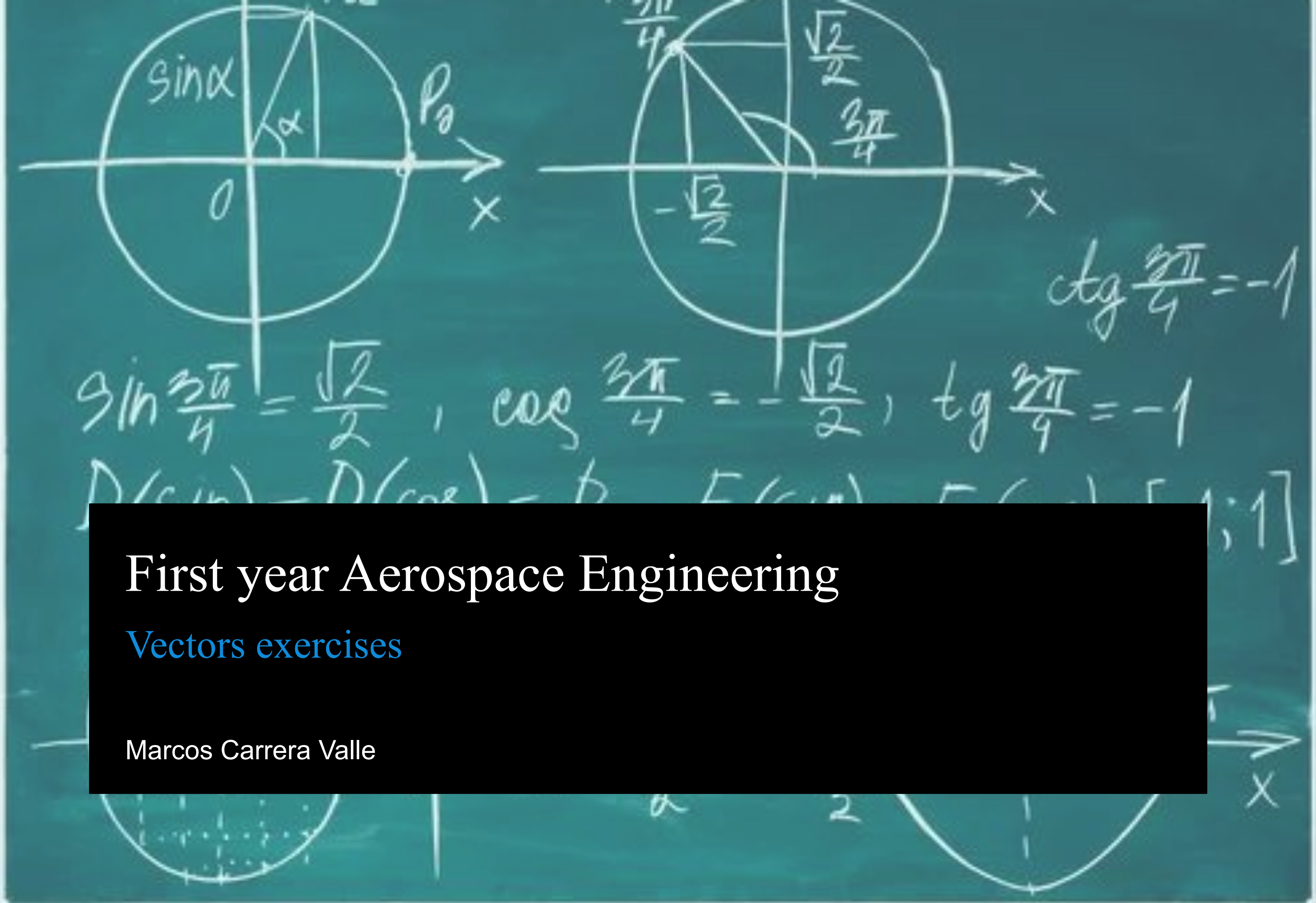

First year Aerospace Engineering

Vectors exercises

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1) Given vectors $a = (2, 3)$ and $b = (4, 1)$, find the vector $a + b$.
2) Given vectors $u = (-5, 2)$ and $v = (3, 4)$, find the vector $u - v$.
3) Given vector $p = (1, -2, 3)$, find the vector $3p$.
4) Given vectors $a = (2, 0, -1)$ and $b = (1, 4, 6)$, find the vector $2a + b$.
5) Given vector $x = (5, -3)$, find the vector $-2x$.
6) Given vectors $c = (3, 4)$ and $d = (-2, 1)$, find the vector $4c - 3d$.
7) If $v = 3i - 2j$ and $w = -2i + 5j$ (where i and j are unit vectors), find the vector $v + w$.
8) Given vector $m = (4, -2, 8)$, find the vector $0.5m$.
9) Given vectors $u = (2, -1, 0)$, $v = (-1, 0, 4)$, and $w = (0, 2, -3)$, find the vector $u - v + w$.
10) Given vectors $a = (1, 2)$ and $b = (-3, 4)$, find the vector $2(a - b)$.

11) Calculate the dot product of vectors a = (3, 5) and b = (-2, 4).
12) Calculate u · v given u = (1, -2, 3) and v = (4, 0, -1).
13) Find the angle (in degrees) between vectors a = (3, 0) and b = (3, 3).
14) Determine if the vectors p = (2, 5, 1) and q = (-3, 1, 1) are perpendicular
15) Find the cross product a × b for vectors a = (1, 3, 4) and b = (2, 7, -5)
16) Find a vector that is perpendicular to both u = (2, 0, 0) and v = (0, 3, 0)
17) Find the area of the parallelogram spanned by the vectors a = (1, 2, 3) and b = (4, 5, 6)
18) Calculate the scalar projection of vector a = (4, 3) onto the direction of vector b = (1, 0)
19) Vectors u = (2, -3) and v = (8, -12) are parallel. Find the scalar k such that v = k*u.
20) Explain why the expression a / b (vector a divided by vector b) is mathematically undefined in standard vector algebra

21) A particle's position vector is given by $r(t) = (3t^2, 4t^3 - 2t)$ meters. Find the speed of the particle at $t = 1$ second.

22) A particle starts from rest ($v = 0$) at $t=0$. Its acceleration is given by $a(t) = (6t, \sin(t))$ m/s^2. Determine the velocity vector at $t = \pi$ seconds.

23) The position of a projectile is given by $r(t) = (15t, 20t - 5t^2)$ meters. Find the time t at which the particle reaches its maximum height (maximum y-coordinate) and determine the velocity vector at that exact moment.

24) A particle moves such that its position is $r(t) = (5\cos(2t), 5\sin(2t))$. Calculate the acceleration vector at $t = \pi/4$ and determine if the magnitude of the acceleration is constant for all t.

25) A particle moves along a path defined by $r(t) = (t^2, 2t - 4)$. Find the time t at which the speed of the particle is at its minimum value.

26) A particle moves such that its position vector is r(t) = (t * sin(t), t * cos(t)). Find the exact magnitude of the acceleration vector at t = pi/2 seconds.

27) A particle has a variable acceleration a(t) = (12t^2 - 4, 6t). At t = 1, its velocity is known to be v(1) = (0, 5). Given that the particle starts at the origin (0,0) at t = 0, find the position vector r(t).

28) A particle moves with a position vector r(t) = (2t, t^2 - 4). Find the time t (where t > 0) when the velocity vector is perpendicular to the position vector.

29) A particle moves along a path defined by r(t) = (t^2 - 2t, 4t). Find the time t at which the speed of the particle is at its absolute minimum, and calculate that minimum speed.

30) Particle A starts at position (2, 5) and moves with a constant velocity of (3, -1). Particle B starts at position (10, -3) and moves with a constant velocity v. If the two particles collide exactly at t = 2 seconds, find the velocity vector v of Particle B and its speed.

36) A rocket launches from the origin with a flight path described by the linear equation $y = 2.4x$. The rocket is travelling at a constant speed of 2600 m/s. Write the vector equation $r = tb$ for this trajectory, where 'b' represents the exact velocity vector.

37) A geostationary satellite transfer vehicle moves along the vector path $r = (50, 0) + t(4, 3)$. A piece of space debris is drifting along a linear path described by $x + 2y = 150$. Find the exact coordinate point where the vehicle's path intersects the debris path.

38) A Mars rover is travelling along a ridge defined by the vector equation $r = (10, 5) + t(3, -1)$. The base station is located at the point (20, 15). The rover needs to turn and travel in a straight line that is perpendicular to its current path to reach the base station. Find the Cartesian equation ($ax + by = c$) of this new path.

39) A solar sail probe is travelling on a vector path r = t(1, sqrt(3)). It approaches a boundary of distinct solar wind pressure marked by the line $y = x$. Find the acute angle (in degrees) between the probe's path and this boundary line.

40) An astronomer maps the path of an asteroid using the linear equation $3x - 4y = 500$ (distance in Mm). To analyze the closest approach to Earth (located at the origin), converting this to vector form is easier.
a) Convert the linear equation to a vector equation $r = a + tb$.
b) Use the dot product property (r dot b = 0) with your new vector equation to find the position vector of the point on the line closest to the origin.

41-45)
A B2 bomber takes off from Airport A (0, 0) and flies towards Airport C (367.5, 490) (coordinates in km).

Unlike a commercial flight, the B2 accelerates constantly such that its velocity vector at time t (hours) is given by:
$v(t) = (60t, 80t)$ km/h

The aircraft consumes fuel based on the square of its speed: Fuel Flow (kg/h) = $0.01 * v \cdot v$.

The B2 takes off with 900 kg of fuel.

a) Determine the time t (in hours) at which the B2 will run out of fuel.

41-45)

When the B2 has exactly 1 hour of fuel remaining (based on the time calculated in part a), the pilot signals Airport B, located at (227.5, 220), for mid-air refueling.

A fuel tanker immediately launches from Airport B.

The tanker flies in a straight line to meet the B2 at point M, which is the point on the B2's flight path that is the shortest physical distance from Airport B.

The tanker adjusts its speed to arrive at point M at the exact same time the B2 arrives there.

Upon meeting, the tanker instantly refills the B2's tank back to 900 kg.

b) Determine the constant velocity vector v(t) the fuel tanker must maintain to achieve this interception.

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After refueling, the B2 continues its journey to Airport C (maintaining its velocity profile v(t)).

c) At what time t does it reach Airport C?

41-45)
After refueling the B2 at point M, the tanker flies to Airport D, located at (187.5, 300).

d) Calculate the total distance traveled by the fuel tanker.

e) The fuel tanker consumes 2 kg of fuel per km traveled. Calculate the total fuel load the tanker needed at takeoff

46-50)

A research rocket launches from a platform at the origin (0, 0, 0). It ascends with a constant acceleration vector a = (0, 30, 40) m/s².

The rocket consumes fuel at a rate proportional to its speed:
Fuel Flow (kg/s) = 4 * |v(t)| (where |v(t)| is the speed in m/s).

The rocket launches with 360,000 kg of fuel on board.

a) Determine the time $t_{burnout}$ (in seconds) at which the rocket will run out of fuel.

46-50)
At $t = 40$ seconds after launch, the rocket ejects a "Data Pod".

The Pod travels in a straight line maintaining the exact velocity vector the rocket had at the moment of ejection.

A recovery drone is stationed on a hovering airship located at coordinates S = (5000, 60000, 80000) (coordinates in meters).

The drone launches from the airship at the exact moment the Pod is ejected ($t=40$).

The drone flies in a straight line to intercept the Pod at the point on the Pod's path that is the shortest distance from the airship.

The drone must arrive at that point at the exact same time the Pod does.

b) What constant velocity vector v_{drone} must the drone have to successfully intercept the Pod?

c) What are the (x, y, z) coordinates where the interception takes place?

46-50)

After collecting the Pod, the drone immediately flies back to the airship at the same speed it arrived with.

d) What is the total distance traveled by the drone?

The drone consumes battery power equivalent to 0.5 units per meter traveled.

e) How many units of power must the drone have available for the entire round trip?

SOLUTIONS

1) (6, 4)
2) (-8, -2)
3) (3, -6, 9)
4) (5, 4, 4)
5) (-10, 6)
6) (18, 13)
7) i + 3j (or (1, 3))
8) (2, -1, 4)
9) (3, 1, -7)
10) (8, -4)

Find the value of each function

11) 14: $(3\times-2) + (5\times4) = -6 + 20 = 14$

12) 1: $(1\times4) + (-2\times0) + (3\times-1) = 4 + 0 - 3 = 1$

13) 45 degrees: $\cos(\theta) = (a\cdot b) / (|a||b|) = 9 / (3 \times \sqrt{18}) = 1/\sqrt{2}$

14) Yes, they are perpendicular: Dot product: $(2\times-3) + (5\times1) + (1\times1) = -6 + 5 + 1 = 0$

15) (-43, 13, 1): $i(-15-28) - j(-5-8) + k(7-6)$

16) (0, 0, 6) or (0, 0, k): Cross product of x-axis and y-axis vectors gives the z-axis.

17) Approx 5.92 (or $\sqrt{35}$): Cross product is (-3, 6, -3). Magnitude is $\sqrt{(9+36+9)} = \sqrt{54} \approx 7.35$. Wait, let me re-calculate: $i(12-15) - j(6-12) + k(5-8) = (-3, 6, -3)$. Magnitude = $\sqrt{(9+36+9)} = \sqrt{54}$.

18) 4: $(a\cdot b) / |b| = 4 / 1 = 4$

19) k = 4: (8/2 = 4 and -12/-3 = 4)

20) Undefined: There is no unique operation for vector division because the cross product and dot product are not reversible functions (many different vectors can produce the same product).

21) 11.66 m/s

22) (3 * pi^2, 2) m/s

23) Time = 2 s; Velocity = (15, 0) m/s

24) a = (0, -20); Yes, the magnitude is constant at 20.

25) t = 0 s

26) Sqrt(4 + pi^2 / 4)

27) r(t) = (t^4 - 2t^2 + t, t^3 + 2t)

28) t = Sqrt(2)

29) Time t = 1; Minimum Speed = 4

30) Velocity v = (-1, 3); Speed = Sqrt(10)

31) r = (0, 5) + t(1, 2)

32) 3x + 4y = 500

33) r = (8, 0) + t(4, 3) (Other answers possible depending on point chosen)

34) y = -0.05x + 4000

35) r = (0, 10) + t(400, 300)

36) r = t(1000, 2400)

37) Intersection Point: (70, 15)

38) 3x - y = 45

39) 15 degrees

40) a) r = (0, -125) + t(4, 3); b) Closest Point: (60, -80)

41-45)
a) 3 hours
b) (-80, 60) km/h
c) 3.5 hours
d) 100 km
e) 720.83 kg

46-50)
a) 60 sec
b) (-500/3, 0, 0)
c) (0, 60000, 80000)
d) 10,000 meters
e) 5,000 units